

Mini-case 3.1.

$$\text{var}(\tilde{r}) = E[(\tilde{r} - \bar{r})^2]$$

$$\text{var}(\tilde{r}) = E(\tilde{r}^2) - \bar{r}^2$$

a) Show that $E[(\tilde{r} - \bar{r})^2] = E(\tilde{r}^2) - \bar{r}^2$

Proof idea: Rearrange the first definition. Then apply properties of expected return

□

$$1. \text{var}(\tilde{r}) = E[(\tilde{r} - \bar{r})^2] = E(\tilde{r}^2 - 2\tilde{r}\bar{r} + \bar{r}^2)$$

2. Considering properties $E[\sum_{i=1}^N \tilde{r}_i^2] = \sum_{i=1}^N E(\tilde{r}_i^2)$ and $E[x \cdot \tilde{r} + c] = x \cdot E(\tilde{r}) + c$ we find:

$$\begin{aligned} E(\tilde{r}^2 - 2\tilde{r}\bar{r} + \bar{r}^2) &= E(\tilde{r}^2) - 2E(\tilde{r}\bar{r}) + E(\bar{r}^2) = E(\tilde{r}^2) - 2\bar{r}\bar{r} + \bar{r}^2 \\ &= E(\tilde{r}^2) - 2\bar{r}^2 + \bar{r}^2 = E(\tilde{r}^2) - \bar{r}^2 \end{aligned}$$

□

b) 1. $\text{var}(A) = E[(\tilde{r} - \bar{r})^2] = 0,25 \cdot (0,2 - 0,05125)^2 + 0,65 \cdot (0,05 -$

$$\bar{r} = 0,2 \cdot 0,25 + 0,05 \cdot 0,65 + 0,1 \cdot (-0,3125) = 0,05125$$

$$- 0,05125)^2 + 0,1 \cdot (-0,3125 - 0,05125)^2 = 0,018764063$$

2. $\text{var}(A) = E(\tilde{r}^2) - \bar{r}^2 = 0,02139 - 0,002626 = 0,018764063$

$$E(\tilde{r}^2) = 0,2^2 \cdot 0,25 + 0,05^2 \cdot 0,65 + 0,1 \cdot (-0,3125)^2 = 0,02139$$

$$\bar{r}^2 = 0,05125^2 = 0,002626$$

Mid-case 3.2.

$$\text{var}(\tilde{r}) = \sigma^2 = \mathbb{E}(\tilde{r} - \bar{r})^2$$

(a)

$$\text{var}(x_1 \tilde{r}_1 + x_2 \tilde{r}_2) = \mathbb{E} \left[\left(x_1 \tilde{r}_1 + x_2 \tilde{r}_2 - \mathbb{E}(x_1 \tilde{r}_1 + x_2 \tilde{r}_2) \right)^2 \right]$$

$$\text{var}(x \tilde{r}_1 + (1-x) \tilde{r}_2) = \mathbb{E} \left[\left(x \tilde{r}_1 + (1-x) \tilde{r}_2 - \mathbb{E}(x \tilde{r}_1 + (1-x) \tilde{r}_2) \right)^2 \right] =$$

$$= \mathbb{E} \left[\left(x \tilde{r}_1 + (1-x) \tilde{r}_2 - (x \bar{r}_1 + (1-x) \bar{r}_2) \right)^2 \right] =$$

$$= \mathbb{E} \left[\left(x \tilde{r}_1 + (1-x) \tilde{r}_2 - x \bar{r}_1 - (1-x) \bar{r}_2 \right)^2 \right] =$$

$$= \mathbb{E} \left[\left(x(\tilde{r}_1 - \bar{r}_1) + (1-x)(\tilde{r}_2 - \bar{r}_2) \right)^2 \right] =$$

$$= \mathbb{E} \left(x^2 (\tilde{r}_1 - \bar{r}_1)^2 + (1-x)^2 (\tilde{r}_2 - \bar{r}_2)^2 + 2x(1-x)(\tilde{r}_1 - \bar{r}_1)(\tilde{r}_2 - \bar{r}_2) \right) =$$

$$= x^2 \mathbb{E}[(\tilde{r}_1 - \bar{r}_1)^2] + (1-x)^2 \mathbb{E}[(\tilde{r}_2 - \bar{r}_2)^2] + 2x(1-x) \mathbb{E}[(\tilde{r}_1 - \bar{r}_1)(\tilde{r}_2 - \bar{r}_2)]$$

$$= x^2 \sigma_1^2 + (1-x)^2 \sigma_2^2 + 2x(1-x) \sigma_{12} =$$

$$\rho_{12} = \frac{\sigma_{12}}{\sigma_1 \sigma_2} \quad | \cdot \sigma_1 \sigma_2$$

$$\sigma_{12} = \rho_{12} \sigma_1 \sigma_2$$

$$= x^2 \sigma_1^2 + (1-x)^2 \sigma_2^2 + 2x(1-x) \rho_{12} \sigma_1 \sigma_2$$

$$\begin{aligned}
 \textcircled{b} \quad 1. \quad \frac{d\sigma^2}{dx} &= [X^2\sigma_1^2 + (1-X)^2\sigma_2^2 + 2X(1-X)\rho_{12}\sigma_1\sigma_2]' = \\
 &= [X^2\sigma_1^2 + (1-X)^2\sigma_2^2 + 2X\rho_{12}\sigma_1\sigma_2 - 2X^2\rho_{12}\sigma_1\sigma_2]' = \\
 &= [X^2\sigma_1^2 + (1-2X+X^2)\sigma_2^2 + 2X\rho_{12}\sigma_1\sigma_2 - 2X^2\rho_{12}\sigma_1\sigma_2]' = \\
 &= X^2\sigma_1^2 - 2\sigma_2^2 + 2X\sigma_2^2 + 2\rho_{12}\sigma_1\sigma_2 - 4X\rho_{12}\sigma_1\sigma_2
 \end{aligned}$$

$$11. \quad 2X\sigma_1^2 - 2\sigma_2^2 + 2X\sigma_2^2 + 2\rho_{12}\sigma_1\sigma_2 - 4X\rho_{12}\sigma_1\sigma_2 = 0$$

$$2X\sigma_1^2 + 2X\sigma_2^2 + 4X\rho_{12}\sigma_1\sigma_2 = -2\rho_{12}\sigma_1\sigma_2 + 2\sigma_2^2$$

$$X(2\sigma_1^2 + 2\sigma_2^2 + 4\rho_{12}\sigma_1\sigma_2) = 2\sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2$$

$$X = \frac{2\sigma_2^2 - 2\rho_{12}\sigma_1\sigma_2}{2\sigma_1^2 + 2\sigma_2^2 + 4\rho_{12}\sigma_1\sigma_2}$$

$$X = \frac{2(\sigma_2^2 - \rho_{12}\sigma_1\sigma_2)}{2\sigma_1^2 + 2\sigma_2^2 + 4\rho_{12}\sigma_1\sigma_2}$$

$$X = \frac{\sigma_2^2 - \rho_{12}\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 + 2\rho_{12}\sigma_1\sigma_2}$$

Mini-case 3.3

$$\textcircled{a} E(\tilde{R}_p) = x_1 E(\tilde{r}_1) + x_2 E(\tilde{r}_2) = x_1 \bar{r}_1 + x_2 \bar{r}_2 = \\ = 0,5 \cdot 0,1314 + 0,5 \cdot 0,0426 = 0,087 = 8,7\%$$

⑥

$$I. x_1 \cdot 0,1314 + x_2 \cdot 0,0426 = 0,09$$

$$x_1 \cdot 0,1314 + (1-x_1) \cdot 0,0426 = 0,09$$

$$x_1 \cdot 0,1314 + 0,0426 - 0,0426 x_1 = 0,09$$

$$0,0888 x_1 = 0,09 - 0,0426$$

$$0,0888 x_1 = 0,0474 \quad | : 0,0888$$

$$x_1 = 0,53378 \dots \approx 53,38\%$$

$$II. x_2 = 1 - x_1 = 1 - 0,53378 \dots \approx 0,466216 \dots \approx 46,62\%$$

$$\textcircled{c} E(\tilde{R}_p) = 0,25 \cdot E(\tilde{r}_g) + 0,25 \cdot E(\tilde{r}_{st}) + 0,25 \cdot E(\tilde{r}_r) + 0,25 \cdot E(\tilde{r}_b) =$$

g - gold

st - stocks

r - REIT - real estate investment trust

b - bonds

$$= 0,25 \cdot 0,1314 + 0,25 \cdot 0,1363 + 0,25 \cdot 0,0426 + 0,25 \cdot 0,0124 =$$

$$= 0,080675 \approx 8,07\%$$

No, this portfolio would have a lower expected return

Mini-case 3.4.

a)

$$X_g = 0,7 = 70\%; \quad X_{st} = 0,3 = 30\%$$

$$\text{Var}(\tilde{R}_p) = X_g^2 \sigma_g^2 + 2 X_g X_{st} \rho_{gst} \sigma_g \sigma_{st} + X_{st}^2 \sigma_{st}^2 =$$

$$\sigma_g^2 = 0,1399^2 = 0,01957201$$

$$\sigma_{st}^2 = 0,1528^2 = 0,02334784$$

$$= 0,7^2 \cdot 0,01957201 + 2 \cdot 0,7 \cdot 0,3 \cdot 0,1 \cdot 0,1399 \cdot 0,1528 + 0,3^2 \cdot 0,02334784 =$$

$$= 0,012589413$$

$$\sigma(\tilde{R}_p) = \sqrt{\text{Var}(\tilde{R}_p)} = 0,112202553 \approx 0,1122$$

b)

$$\rho_{BP} = \frac{\sigma_{BP}}{\sigma_B \sigma_P} = \frac{0,002666828}{0,0538 \cdot 0,112202553} = 0,441784065$$

$$\sigma_P = 0,112202553 \approx 0,1122$$

$$\sigma_B = 0,0538$$

$$\sigma_{BP} = \sum_{i=1}^2 X_i \sigma_{iB} = X_1 \sigma_{1B} + X_2 \sigma_{2B} =$$

$$\sigma_{1B} = \rho_{1B} \sigma_1 \sigma_B = 0,45 \cdot 0,1399 \cdot 0,0538 = 0,003386979$$

$$\rho_{1B} = \frac{\sigma_{1B}}{\sigma_1 \sigma_B}$$

$$\sigma_{2B} = \rho_{2B} \sigma_2 \sigma_B = 0,12 \cdot 0,1528 \cdot 0,0538 = 0,000986477$$

$$= 0,7 \cdot \sigma_{1B} + 0,3 \cdot \sigma_{2B} = 0,7 \cdot 0,003386979 + 0,3 \cdot 0,000986477 =$$

$$= 0,002666828$$

c)

$$\sigma_p^2 = X^T \Sigma X = (X_g \quad X_{st} \quad X_B) \begin{pmatrix} \sigma_g^2 & \sigma_{gst} & \sigma_{gB} \\ \sigma_{sg} & \sigma_{st}^2 & \sigma_{stB} \\ \sigma_{Bg} & \sigma_{Bst} & \sigma_B^2 \end{pmatrix} \begin{pmatrix} X_g \\ X_{st} \\ X_B \end{pmatrix}$$

$$\sigma_{gst} = \rho_{gst} \sigma_g \sigma_{st} = 0,1 \cdot 0,1399 \cdot 0,1528 = 0,002137672$$

$$\sigma_{gB} = \rho_{gB} \sigma_g \sigma_B = 0,45 \cdot 0,1399 \cdot 0,0538 = 0,003386979$$

$$\sigma_{stB} = \rho_{stB} \sigma_{st} \sigma_B = 0,12 \cdot 0,1528 \cdot 0,0538 = 0,000986477$$

$$\sigma_g^2 = 0,1399^2 = 0,01957201$$

$$\sigma_{st}^2 = 0,1528^2 = 0,02334784$$

$$\sigma_B^2 = 0,0538^2 = 0,00289444$$

Let Σ be A , then

$$\sigma_p^2 = x^T A x = (0,33 \quad 0,33 \quad 0,33) A \begin{pmatrix} 0,33 \\ 0,33 \\ 0,33 \end{pmatrix} =$$

$$= \begin{pmatrix} 0,008365554 & 0,008823996 & 0,002422632 \end{pmatrix} \begin{pmatrix} 0,33 \\ 0,33 \\ 0,33 \end{pmatrix} =$$

$$= 0,006537394$$

$$\sigma_p = 0,080854153 \approx 0,0809$$

Yes, the standard deviation would be considerably lower, which means that bonds provide diversification benefits into portfolio.